$$\delta(q_0, a, a) = (q_0, aa), (q_1, \varepsilon)$$
$$\delta(q_0, b, b) = (q_0, bb), (q_1, \varepsilon)$$

So, the first condition fails. To be deterministic, both the conditions shown in the definition should be satisfied. Since one of the conditions is not met, the PDA is *non-deterministic*.

5.7. CFG to PDA

It is quite easy to get a PDA from the context free grammar. This is possible only from a CFG which is in GNF. So, given any grammar, first obtain the grammar in GNF and then obtain the PDA. The steps to be followed to convert a grammar to its equivalent PDA are shown below:

1. Convert the grammar into GNF
2. Let q_0 be the start state and Z_0 is the initial symbol on the stack. Without consuming any input, push the start symbol S onto the stack and change the state to q_1. The transition for this can be

   $\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$
3. For each production of the form

   $A \rightarrow a\alpha$

   introduce the transition

   $\delta(q_1, a, A) = (q_1, \alpha)$
4. Finally, in state q_1, without consuming any input, change the state to q_f which is an accepting state. The transition for this can be of the form

   $\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

Example 5.18: For the below grammar

$$S \rightarrow aABC$$
$$A \rightarrow aB|a$$
$$B \rightarrow bA|b$$
$$C \rightarrow a$$

obtain the corresponding PDA.

Let q_0 be the start state and Z_0 the initial symbol on the stack.

Step 1: Push the start symbol S on to the stack and change the state to q_1. The transition for this can be of the form

$$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$$

Step 2: For each production $A \rightarrow a\alpha$ introduce the transition

$$\delta(q_1, a, A) = (q_1, \alpha)$$

This can be done as shown below:

Production			Transition		
S	$\rightarrow$	aABC	$\delta(q_1, a, S)$	=	(q_1, ABC)
A	$\rightarrow$	aB	$\delta(q_1, a, A)$	=	(q_1, B)
A	$\rightarrow$	a	$\delta(q_1, a, A)$	=	(q_1, ε)
B	$\rightarrow$	bA	$\delta(q_1, b, B)$	=	(q_1, A)
B	$\rightarrow$	b	$\delta(q_1, b, B)$	=	(q_1, ε)
C	$\rightarrow$	a	$\delta(q_1, a, C)$	=	(q_1, ε)

Step 3: Finally in state q_1, without consuming any input change the state to q_f which is an accepting state, i.e.,

$$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$$

So, the PDA M is given by $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ where

$Q = \{q_0, q_1, q_f\}$
$\Sigma = \{a, b\}$
$\Gamma = \{S, A, B, C, Z_0\}$

δ : is shown below:

$$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$$
$$\delta(q_1, a, S) = (q_1, ABC)$$
$$\delta(q_1, a, A) = (q_1, B)$$
$$\delta(q_1, a, A) = (q_1, \varepsilon)$$
$$\delta(q_1, b, B) = (q_1, A)$$
$$\delta(q_1, b, B) = (q_1, \varepsilon)$$
$$\delta(q_1, a, C) = (q_1, \varepsilon)$$
$$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$$

$q_0 \in Q$ is the start state of machine.

$Z_0 \in \Gamma$ is the initial symbol on the stack.

$F = \{q_f\}$ is the final state.

Note that the terminals of grammar G will be the input symbols in PDA and the non-terminals will be the stack symbols in PDA.The derivation from the grammar is shown below:

$$\begin{aligned} S &\Rightarrow aABC \\ &\Rightarrow aaBBC \\ &\Rightarrow aabBC \\ &\Rightarrow aabbC \\ &\Rightarrow aabba \end{aligned}$$

The string aabba is derived from the start symbol S. The same string should be accepted by PDA also. The moves made by the PDA are shown below:

Initial ID

$$\begin{aligned} (q_0, aabba, Z_0) &\vdash (q_1, aabba, SZ_0) && \text{By Rule 1} \\ &\vdash (q_1, abba, ABCZ_0) && \text{By Rule 2} \\ &\vdash (q_1, bba,, BBCZ_0) && \text{By Rule 3} \\ &\vdash (q_1, ba, BCZ_0) && \text{By Rule 6} \\ &\vdash (q_1, a, CZ_0) && \text{By Rule 6} \\ &\vdash (q_1, \varepsilon, Z_0) && \text{By Rule 7} \\ &\quad (q_f, \varepsilon, Z_0) && \text{By Rule 8 (Final configuration)} \end{aligned}$$

Since q_f is the final state and input string is ε in the final configuration, the string

aabba

is accepted by the PDA.

Example 5.19: For the grammar

$$\begin{aligned} S &\rightarrow aABB \mid aAA \\ A &\rightarrow aBB \mid a \end{aligned}$$

$$B \rightarrow bBB \mid A$$
$$C \rightarrow a$$

obtain the corresponding PDA.

Note: To obtain a PDA from CFG, the grammar should be in GNF. All the productions except the production

$$B \rightarrow A$$

are in GNF. Since one of the production is not in GNF, the given grammar is not in GNF. This can be easily converted into GNF. Note that all A productions are in GNF. So, by substituting for A in the above production we get B-productions also in GNF as shown below:

$$B \rightarrow bBB \mid aBB \mid a$$

Now, the new grammar which is in GNF can take the form:

$$S \rightarrow aABB \mid aAA$$
$$A \rightarrow aBB \mid a$$
$$B \rightarrow bBB \mid aBB \mid a$$
$$C \rightarrow a$$

Now, the CFG can be converted into PDA. Let q_0 be the start state and Z_0 the initial symbol on the stack.

Step 1: Push the start symbol S on to the stack and change the state to q_1. The transition for this can be of the form

$$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$$

Step 2: For each production $A \rightarrow a\alpha$ introduce the transition

$$\delta(q_1, a, A) = (q_1, \alpha)$$

The following table shows how the various transtitions are obtained from the productions from the CFG.

Production			Transition		
S	$\rightarrow$	aABB	$\delta(q_1, a, S)$	=	(q_1, ABB)
S	$\rightarrow$	aAA	$\delta(q_1, a, S)$	=	(q_1, AA)
A	$\rightarrow$	aBB	$\delta(q_1, a, A)$	=	(q_1, BB)
A	$\rightarrow$	a	$\delta(q_1, a, A)$	=	(q_1, ε)
B	$\rightarrow$	bBB	$\delta(q_1, b, B)$	=	(q_1, BB)
B	$\rightarrow$	aBB	$\delta(q_1, a, B)$	=	(q_1, BB)
B	$\rightarrow$	a	$\delta(q_1, a, B)$	=	(q_1, ε)
C	$\rightarrow$	a	$\delta(q_1, a, C)$	=	(q_1, ε)

Step 3: Finally in state q_1, without consuming any input change the state to q_f which is an accepting state, i.e.,

$$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$$

So, the PDA M is given by $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

where

$Q = \{q_0, q_1, q_f\}$
$\Sigma = \{a, b\}$
$\Gamma = \{S, A, B, C, Z_0\}$

δ : is shown below:

$$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$$
$$\delta(q_1, a, S) = (q_1, ABB)$$
$$\delta(q_1, a, S) = (q_1, AA)$$
$$\delta(q_1, a, A) = (q_1, BB)$$
$$\delta(q_1, a, A) = (q_1, \varepsilon)$$
$$\delta(q_1, b, B) = (q_1, BB)$$
$$\delta(q_1, a, B) = (q_1, BB)$$
$$\delta(q_1, a, B) = (q_1, \varepsilon)$$
$$\delta(q_1, a, C) = (q_1, \varepsilon)$$
$$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$$

$q_0 \in Q$ is the start state of machine.